

## **ENEL700**

### **Lecture 10**

#### Exercise 1:

Calculate the maximum channel capacity of a voice-grade telephone line with a bandwidth of 3100 Hz and an S/N of 30 dB.

#### Exercise 2:

The bandwidth of a communication channel is 12.5 kHz. The S/N ratio is 25dB. Calculate

- (a) The maximum theoretical data rate in bits per second
- (b) The maximum theoretical channel capacity
- (c) The number of coding levels  $N$  needed to achieve the maximum speed.

#### Exercise 3:

If 5 errors occurred in 1.0 second interval of a 10Mbps communication channel transmission, calculate the BER.

#### Exercise 4:

For ASCII characters represented by “1011001” and “0010100” bit streams, find the the characters transmitted including parity bit in Odd and Even parity error detection systems.

#### Exercise 5:

For a given message function of 1101001 and a preselected constant 101, find out the Cyclical Redundancy Check (CRC) character to be appended to the transmitted data.

Exercise 6:

Assume an 8-bit character 01101010 is transmitted with 4 Hamming bits placed at the locations indicated (H) below.

|    |    |    |   |   |   |   |   |   |   |   |   |
|----|----|----|---|---|---|---|---|---|---|---|---|
| 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| H  | 0  | 1  | H | 1 | 0 | H | 1 | 0 | H | 1 | 0 |

- (a) Prove that 4 Hamming bits are sufficient for the data character
- (b) Compute the Hamming code bits
- (c) Show the complete 12-bit transmitted word